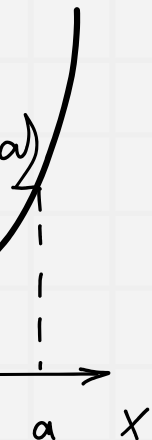


$$\int_0^1 dy \int_0^{1-x} f(x) dx + \int_0^{1/\sqrt{2}} dy \int_0^{f(2x)} f(2x) dx =$$

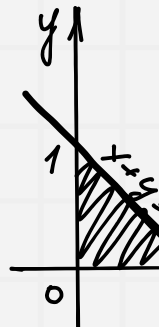
$$2\sqrt{y^2 - x^2}$$



Números Binários

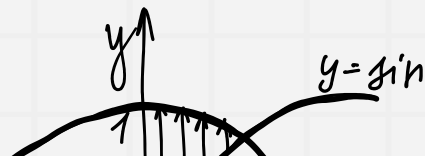
Matemática Computacional Aplicada

Daniel Henrique Matos de Paiva



$$\begin{aligned} X &= 2y^2 \\ Z &= 1 + \sqrt{y} \\ Z &= 4 + \sqrt{y} \end{aligned}$$

$$\int_0^1 dx \int_0^{1-x} x^2 z^{10(x+3y)} du =$$



$$\int_0^{1-x} \int_0^{10(x+3y)} \int_0^{10(x+3y)} x^2 dz =$$

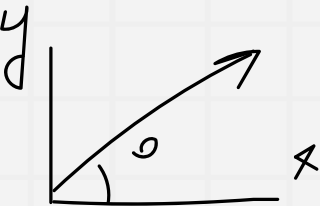
$$\frac{3, x=5}{2+4y^2} \frac{2+4y^2}{2+4y^2}$$

$$V: z=10(x+3y), x+y+z=10, x=0, y=0, z=0$$

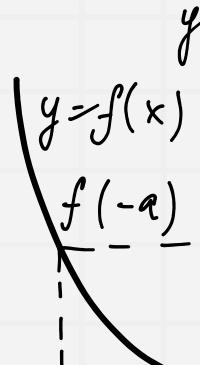
01

Subtracao

Como funciona?



$$= \int_0^1$$



fdy

Regras

- Método "base 2"
- Método "vem -1"



$$5 = 2$$

$$\begin{array}{r}
 \begin{array}{cc}
 \overset{4}{\cancel{5}} & \overset{10}{\cancel{0}}
 \end{array}
 \begin{array}{l}
 (10) \\
 (10)
 \end{array} \\
 \begin{array}{cc}
 \text{— } 2 & 5
 \end{array}
 \begin{array}{l}
 (10) \\
 (10)
 \end{array} \\
 \hline
 \begin{array}{cc}
 2 & 5
 \end{array}
 \begin{array}{l}
 \\
 (10)
 \end{array}
 \end{array}$$

^r
fdy

Regras

- Método "base 2"

- Método "vem -1"



$$5 = 2$$

fdy

Regras – Base 2

- $0 - 0 = 0$
- $1 - 1 = 0$
- $1 - 0 = 1$
- $0 - 1 = 1_{(2)}$

" $10_{(2)} - 1_{(2)} = 1_{(2)}$ " " $2_{(10)} - 1_{(10)} = 1_{(10)}$ "

- Se for impossível subtrair os valores,
pega 1 emprestado "ao vizinho"



$5 = 2$

fdy

Regras – Base 2

$$11_{(2)} - 10_{(2)} = ?$$



$$5 = 2$$

		1	1		
	-	1	0		
		0	1		

fdy

Regras – Base 2

$$11_{(2)} - 10_{(2)} = 1_{(2)}$$



$$5 = 2$$

fdy

Regras – Base 2

$$10_{(2)} - 1_{(2)} = ?$$



$$5 = 2$$

0

2

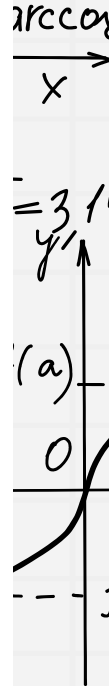
~~1~~ 0

-

1

0 1

$$\begin{array}{r}
 \overset{0}{x} \overset{2}{10} \underset{(2)}{1} \\
 - \quad \quad \quad \underset{(2)}{1} \\
 \hline
 \underset{(2)}{0} \underset{(2)}{1}
 \end{array}$$



fdy

Regras – Base 2

$$110_{(2)} - 11_{(2)} = ?$$



$$5 = 2$$

		0	2	2
	1	1	0	
	-	1	1	
	0	1	1	

fdy

Regras – Base 2

$$1111_{(2)} - 1001_{(2)} = ?$$



$$5 = 2$$

		1	1	1	1	
	-	1	0	0	1	
		0	1	1	0	

fdy

Regras – Base 2

$$11110_{(2)} - 11011_{(2)} =$$

?



$$5 = 2$$

					0	2	2
			1	1	1	1	0
	-		1	1	0	1	1
			0	0	0	1	1

					0	10	10
			1	1	1	1	0
-			1	1	0	1	1
			0	0	0	1	1

fdy

Regras

- Método "base 2"

- Método "vem -1"



$$5 = 2$$

$f dy$

Regras – Vem -1

- $0 - 0 = 0$
 - $1 - 1 = 0$
 - $1 - 0 = 1$
 - $0 - 1 = 1$ e Vem -1
- Se for impossível subtrair os valores,
leva -1 para o vizinho superior



$$5 = 2$$

fdy

Regras – Vem -1

$$110_{(2)} - 11_{(2)} = ?$$



$$5 = 2$$

	1	1	0
	-1	-1	
	0	0	
-		1	1
	0	1	1

fdy

Regras – Vem -1

$$11110_{(2)} - 11011_{(2)} =$$

?



$$5 = 2$$

	1	1	1	1	0
			-1	-1	
-			0	0	
	1	1	0	1	1
	0	0	0	1	1

^r
fdy

Regras – Vem -1

100110₍₂₎ -

10011₍₂₎ = ?



5 = 2

	1	0	0	1	1	0
	-1			-1	-1	
	0			0	0	
-		1	0	0	1	1
		1	0	0	1	1

fdy

Regras – Vem -1

10000000₍₂₎ –

1100111₍₂₎ = ?



5 = 2

	1	0	0	0	0	0	0	0
	1	1	1	1	1	1	1	
	0	1	1	1	1	1	1	
—		1	1	0	0	1	1	1
	0	0	0	1	1	0	0	1

|

^r
f dy

Regras – Vem -1

11000101₍₂₎ –

1100111₍₂₎ = ?

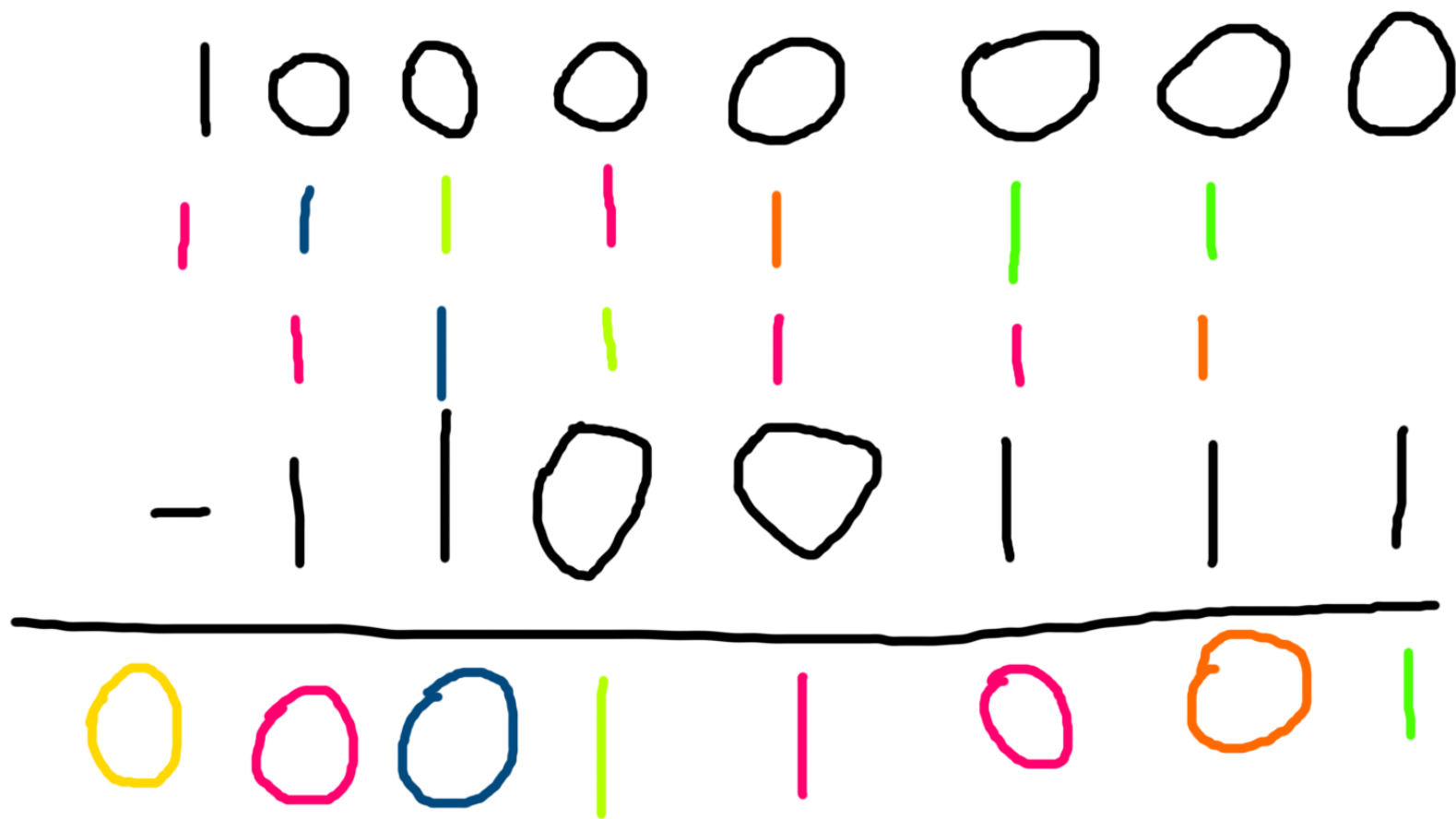


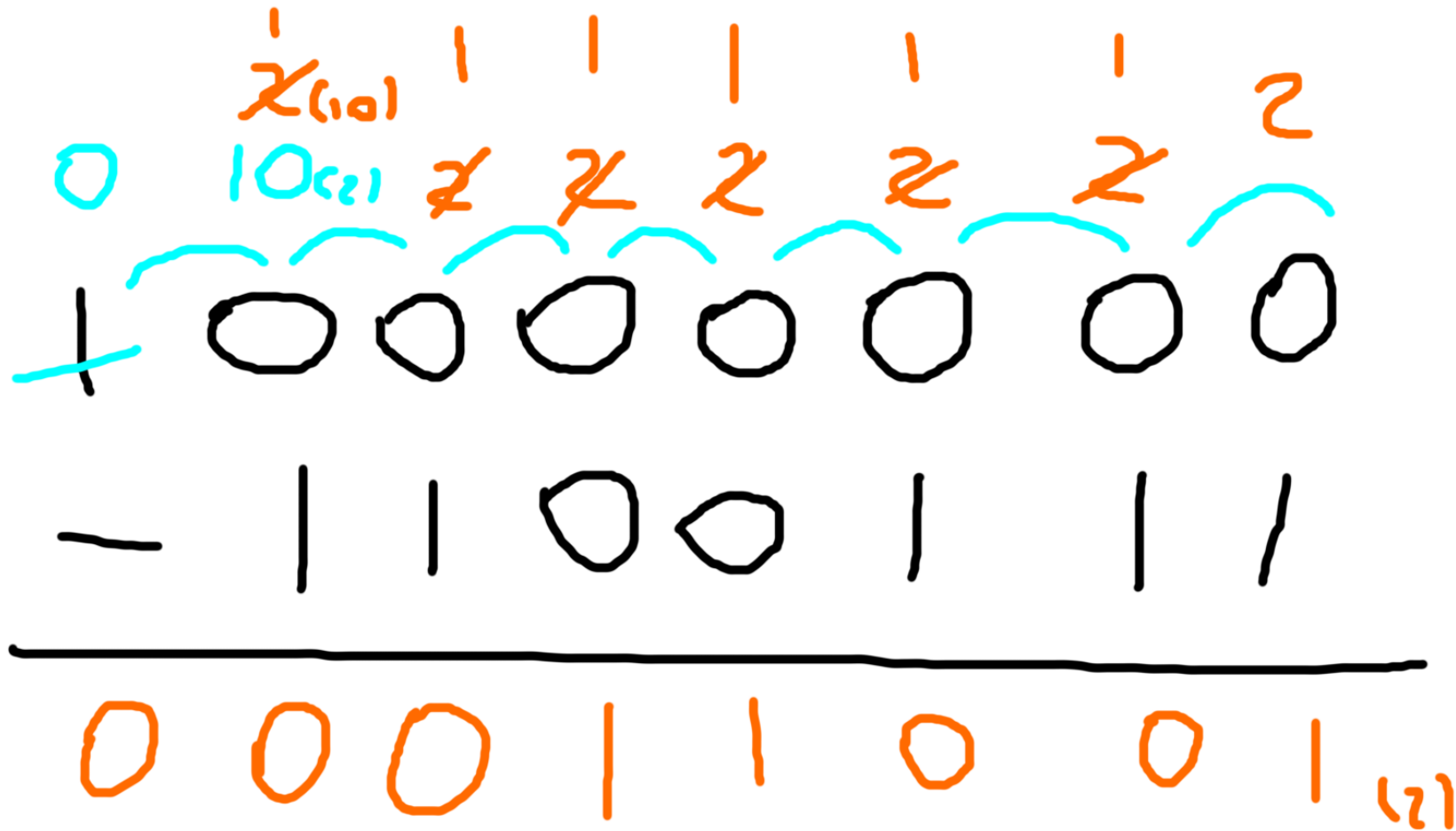
$$5 = 2$$

	1	1	0	0	0	1	0	1
	1	1	1	1	1	1		
	0	0	1	1	1	0		
—		1	1	0	0	1	1	1
		1	0	1	1	1	1	0



$$\begin{array}{r}
 \overset{0}{\times} 10 \overset{2}{\nearrow} \\
 \quad \quad \quad (2) \\
 - \quad \quad \quad | \quad (2) \\
 \hline
 0 \quad | \quad (2)
 \end{array}$$





0	2	1	1	1	2	2	
1	1	2	2	2	0	0	

-	1	1	0	0	1	1	1
---	---	---	---	---	---	---	---

1	0	1	1	1	1	1	0 ₍₂₎
---	---	---	---	---	---	---	------------------